Locomotion for *Walter* (Walking Alternating Tripod for Evolutionary Research): CAD, Firmware, Optimization, and MuJoCo PPO

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Abstract

This report summarizes an Engineering 90 capstone on *Walter*, our twelve-degree-of-freedom hexapod (**W**alking **A**lternating **T**ripod for **E**volutionary **R**esearch). The mechanical structure uses **CAD-designed, 3D-printed** body, leg, and linkage parts assembled with hobby servos. Electronics include a Teensy 4.1, Pololu eighteen-channel servo controller (Maestro family), SparkFun BMI270 IMU, regulated power, and three iterations of custom *Hexapod V2* PCBs. **Proposed work** combined layered, bio-inspired control—tripod locomotion, IMU-driven reflexes, and slow on-device adaptation—with offline gait optimization and reinforcement learning in simulation. **Benefits** include interpretable gaits, repeatable artifacts, safety-aware command bounds, and policies trained without hand-written joint trajectories. **Accomplished:** open-loop optimization to firmware (`hexapod_openloop_sine`); discrete tripod baseline (`hexapod_walking_v2`); MuJoCo [1] PPO [2] via Stable-Baselines3 [3] with released evaluations. **Requirements** (`PHASES.md`) are **partially met:** hardware, CAD/mechanical documentation in this report, open-loop optimization, and RL-in-simulation are evidenced; Phase 1–4 adaptive IMU firmware paths are largely absent from the tree. Source materials are maintained at https://github.com/HowardWHSrun/WALTER. **Cost:** electronics (three BOMs) US$595.20; three PCB orders US$197.47; **grand total US$792.67** (one PCB order awaited payment at writing). Filament and fasteners are outside this sum.

1 Introduction

Reliable legged locomotion requires coordinating periodic limb motion with feedback under limited sensing and computation. Biological systems, in particular insects, organize locomotor control in *layers* that are by now standard in the comparative and robotics literature: central pattern generators (CPGs) and inter-leg coordination establish a rhythmic substrate; fast mechanosensory and

proprioceptive loops modulate timing and force in response to body perturbations; and slower processes adjust parameters over behavioral time scales [4, 5]. This project treats that decomposition not as metaphor but as a **design principle**: each major engineering choice on *Walter* was selected to preserve an explicit correspondence to this biological organization, so that hardware, firmware, and learning algorithms remain interpretable as implementations of the same layered story.

Bio-inspired correspondences. The **alternating tripod** gait used on the platform parallels the phase relationships observed in fast hexapedal locomotion: a fixed alternation of stance groups provides a stable, low-dimensional rhythmic prior before any learning is applied. **Inertial sensing** (a six-axis IMU fused for roll, pitch, and body rates) stands in for the distributed mechanoreception and body-state estimation that insects use to stabilize posture and trigger reflexive adjustments; fusion emphasizes roll and pitch over unreliable magnetic heading, consistent with how legged systems exploit gravito-inertial cues on structured terrain [6, 7]. The intended **fast** control tick (filtering, stability scoring, reflexive modulation of stride variables) and **slow** tick (windowed costs, optional parameter adaptation) mirror the separation between millisecond-scale corrective reflexes and slower tuning of gait parameters documented in adaptive locomotion models [5]. **Reinforcement learning in simulation** (PPO in MuJoCo) is used not to replace that hierarchy but to explore high-dimensional dynamics and shaped rewards while structured policies and phase cues retain a CPG-like inductive bias. **Optimization of a compact open-loop gait law** prior to closing sensor loops follows the same logic: isolate an interpretable rhythmic generator, then add sensory correction, analogous to decoupling pattern generation from sensory modulation in biological models.

Platform and scope. The physical system is *Walter*, a twelve-degree-of-freedom hexapod (WALTER: **W**alking **A**lternating **T**ripod for **E**volutionary **R**esearch). The chassis and leg mechanisms are CAD-designed, additively manufactured, and integrated with commodity servos and a custom *Hexapod V2* printed circuit board (Sec. 6.5–6.6). The target software architecture described in the project documentation specifies tripod-based gait generation, complementary-filter attitude estimation, stability scoring, reflexive modulation of gait variables, and optional slow adaptation of a low-dimensional parameter vector with nonvolatile storage; Sec. 3–5 report the degree to which each layer is realized in the current repository.

Repository. Source materials, including firmware, simulation code, and documentation, are maintained at `https://github.com/HowardWHSrun/WALTER`.

Organization of the technical work. Development proceeds along **three tracks** (Sec. 3–5): **(1)** surrogate gait optimization with deterministic export to embedded open-loop code (*complete*); **(2)** proximal policy optimization in MuJoCo with IMU-inclusive observations and tripod-structured or preset gait priors (*near complete* in simulation); and **(3)** extension of that stack with visual state for distance-aware rewards (*not yet implemented*). Sec. 2 develops the optimization and reinforcement-learning formalism shared across these tracks.

2 Mathematical background: optimization and reinforcement learning

2.1 Reduced-order gait optimization (Track 1)

2.1.1 Decision variables and bounds

The optimizer acts on a vector $\boldsymbol{\phi} \in \mathbb{R}^{25}$ matching `hexapod_no_imu_optimization.py`. Indices are concatenated as

$$\boldsymbol{\phi} = (\boldsymbol{\theta}_{\text{abd,rest}}^{\top}, \boldsymbol{\theta}_{\text{flx,rest}}^{\top}, \boldsymbol{\delta}_{\text{abd}}^{\top}, \boldsymbol{\delta}_{\text{flx}}^{\top}, f)^{\top}, \tag{1}$$

where each of $\boldsymbol{\theta}_{\text{abd,rest}}, \boldsymbol{\theta}_{\text{flx,rest}}, \boldsymbol{\delta}_{\text{abd}}, \boldsymbol{\delta}_{\text{flx}} \in \mathbb{R}^6$ are in *degrees* in the code (converted to radians internally), indexed by leg order FL, FR, ML, MR, RL, RR, and $f \in \mathbb{R}_+$ is the common gait frequency in hertz. Box constraints are implemented as `scipy.optimize.Bounds`: for example $f \in [0.4,\, 4.0]\,\text{Hz}$; resting abduction per joint in $[-25°, 25°]$; resting flexion in $[10°, 65°]$; phase shifts in $[-120°, 120°]$. These define a feasible orthotope $\boldsymbol{\phi}_{\min} \le \boldsymbol{\phi} \le \boldsymbol{\phi}_{\max}$.

2.1.2 Open-loop joint commands (sinusoidal law)

Let $\omega = 2\pi f$. For leg $\ell \in \{1, \ldots, 6\}$ with fixed tripod offset $\psi_\ell \in \{0, \pi\}$ (alternating tripod), time $t \ge 0$, and fixed amplitude tables $\bar{a}_{\text{abd},\ell}$, $\bar{a}_{\text{flx},\ell}$ (defaults in code, not decision variables), the commanded abduction and flexion angles are

$$\phi_{\text{abd},\ell}(t) = \theta_{\text{abd,rest},\ell} + \bar{a}_{\text{abd},\ell}\, \sin\!\big(\omega t + \psi_\ell + \delta_{\text{abd},\ell}\big), \tag{2}$$

$$\phi_{\text{flx},\ell}(t) = \theta_{\text{flx,rest},\ell} + \bar{a}_{\text{flx},\ell}\, \sin\!\big(\omega t + \psi_\ell + \delta_{\text{flx},\ell}\big) - \Delta_{\text{lift},\ell}(t), \tag{3}$$

where $\Delta_{\text{lift},\ell}$ implements swing-phase foot clearance (subtracting flexion when the abduction sine is positive), identical in structure to the firmware `LIFT_SWING_DEG_PEAK` term. Velocities $\dot{\phi}_{\text{abd},\ell}$, $\dot{\phi}_{\text{flx},\ell}$ are obtained by differentiation and feed the planar surrogate dynamics.

2.1.3 Surrogate planar dynamics and simulation

The reduced model integrates an eight-dimensional state

$$\mathbf{x}(t) = (x,\, \dot{x},\, z,\, \dot{z},\, \phi_{\text{roll}},\, \dot{\phi}_{\text{roll}},\, \phi_{\text{pitch}},\, \dot{\phi}_{\text{pitch}})^{\top}, \tag{4}$$

with $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\phi})$ assembled from lumped masses, soft ground contact, leg thrust and support forces derived from $\phi_{\text{abd},\ell}$, $\phi_{\text{flx},\ell}$ and contact weights, and linear damping on body motion and attitude rates (`body_ode` in the script). For fixed $\boldsymbol{\phi}$, the trajectory is obtained by `solve_ivp` (Runge–Kutta) over $t \in [0, T]$ with T the simulation horizon (`sim_duration_s`), sampled at N time points (`sample_count`).

2.1.4 Scalar objective (explicit weighted sum)

Let a subscript "ss" denote statistics evaluated on a *steady-state* suffix of the horizon (the code uses the last 60% of samples for oscillation metrics and mean speed). Write $\bar{v}_x$ for mean forward velocity $\mathbb{E}_{\mathrm{ss}}[\dot{x}]$, and RMS values $\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{roll}}]$, $\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{pitch}}]$, $\mathrm{RMS}_{\mathrm{ss}}[z]$. Let $q_{\mathrm{servo}} \geq 0$ be a slack on peak torque proxy exceeding a threshold (0.85 of stall in the implementation), and $\sigma_{\mathrm{contact}}$ the time mean of the per-timestep standard deviation of contact weights across legs. With nonnegative weights $w_\cdot$ matching `OptimizationConfig`, the implemented cost is

$$\begin{aligned}\mathcal{J}(\boldsymbol{\phi}) = & -w_{\mathrm{fwd}}\,\bar{v}_x + w_{\mathrm{back}}\left[\max(0, -\bar{v}_x)\right]^2 \\ & + w_{\mathrm{rp}}\Big(\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{roll}}]^2 + \mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{pitch}}]^2\Big) + w_z\,\mathrm{RMS}_{\mathrm{ss}}[z]^2 \\ & + w_{\mathrm{servo}}\,q_{\mathrm{servo}}^2 + w_{\mathrm{contact}}\,\sigma_{\mathrm{contact}}^2 + w_{\mathrm{reg}}\,\|\boldsymbol{\phi} - \boldsymbol{\phi}_0\|_{\mathrm{scaled}}^2,\end{aligned} \tag{5}$$

where $\boldsymbol{\phi}_0$ is the default parameter vector and $\|\cdot\|_{\mathrm{scaled}}$ averages coordinatewise squared normalized deviation from the box midpoint (Tikhonov-style regularization keeping $\boldsymbol{\phi}$ in a "reasonable" neighborhood). Optional terms penalize shortfall below a target mean speed if `target_forward_speed_mps` $>$ 0. Infeasible simulations return a large penalty (10^6). Equation (5) is the precise surrogate counterpart of the abstract sum $\sum_k w_k c_k$ used earlier: here each c_k is spelled out.

2.1.5 Optimization algorithm and restarts

The problem solved in software is

$$\min_{\boldsymbol{\phi}} \; \mathcal{J}(\boldsymbol{\phi}) \quad \text{subject to} \quad \boldsymbol{\phi}_{\min} \leq \boldsymbol{\phi} \leq \boldsymbol{\phi}_{\max}. \tag{6}$$

L-BFGS-B (`scipy.optimize.minimize`, method `L-BFGS-B`) approximates curvature of $\mathcal{J}$ with a limited-memory inverse Hessian and projects each iterate onto the box. Each evaluation of $\mathcal{J}$ requires one numerical integration of the surrogate. The outer loop runs R random restarts (`restarts`): one start at $\boldsymbol{\phi}_0$ and $R-1$ uniform samples in the box; the best $\boldsymbol{\phi}^\star$ by lowest $\mathcal{J}$ is retained. Limits `maxiter`, `maxfun`, and `ftol` cap work per restart.

The **code path** from optimization to firmware is deterministic: $\boldsymbol{\phi}^\star$ is serialized to `no_imu_optimization_summary.` `sync_gait_params_from_json.py` emits `optimized_gait_params.h`; `hexapod_openloop_sine.ino` evaluates the same `leg_commands()` map, applies bench degree-to-pulse gains and clamps, and streams servo targets. Thus optimization-to-code is *algebraic parity*, not a learned policy.

2.2 Markov decision processes and returns

Reinforcement learning treats legged control as a **Markov decision process** (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$: states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, transition density $P(s'|s,a)$ from the simulator (MuJoCo), reward

$r(s,a)$, and discount $\gamma \in (0,1)$. A stochastic policy $\pi_\theta(a|s)$ with parameters θ induces trajectories $\tau = (s_0, a_0, r_0, \ldots)$. The **return** from time t is the discounted sum $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}$. The **state-value** and **action-value** functions under π_θ are

$$V^{\pi_\theta}(s) = \mathbb{E}[G_t \mid s_t = s], \qquad Q^{\pi_\theta}(s,a) = \mathbb{E}[G_t \mid s_t = s, a_t = a]. \tag{7}$$

The **advantage** $A^{\pi_\theta}(s,a) = Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s)$ measures how much better action a is than the policy average in state s. Policy-search methods estimate gradients of the expected return $\eta(\theta) = \mathbb{E}[G_0]$; the **policy gradient theorem** motivates updates proportional to $\nabla_\theta \log \pi_\theta(a|s)\, A(s,a)$, but raw Monte Carlo estimates have high variance.

2.3 Generalized advantage estimation (GAE) and the value baseline

Practical actor–critic algorithms maintain a **critic** $V_\psi(s) \approx V^{\pi_\theta}(s)$ with parameters ψ. One-step **TD residuals** are $\delta_t = r_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t)$. **Generalized advantage estimation** (GAE) combines multi-step returns with an exponential weighting by $\lambda \in [0,1]$ (see Schulman *et al.* on GAE; used inside SB3's PPO [2, 3]):

$$\hat{A}_t = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \, \delta_{t+\ell}, \tag{8}$$

which trades bias (relying on V_ψ) against variance (longer rollouts). Stable-Baselines3 uses this form when `gae_lambda` is set in the PPO configuration.

2.4 Proximal policy optimization (PPO)

PPO [2] stabilizes policy updates by limiting how far θ moves from the behavior policy θ_{old} that collected the batch. Define the **probability ratio**

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}. \tag{9}$$

For continuous actions, π_θ is typically Gaussian; r_t is the ratio of densities. The **clipped surrogate objective** for a batch of timesteps is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta), 1-\epsilon, 1+\epsilon\big)\,\hat{A}_t\big)\Big], \tag{10}$$

with clip range $\epsilon \in (0,1)$ (often 0.2). Large policy steps that inflate r_t when $\hat{A}_t > 0$ (or shrink r_t when $\hat{A}_t < 0$) are clipped so the effective objective stops increasing, reducing catastrophic updates common in vanilla policy gradients.

Let $\hat{R}_t$ denote a target return for the critic (e.g., n-step return or GAE-backed value target). With

coefficients $c_v, c_e > 0$, the **composite loss** maximized in practice is the negative of

$$\mathcal{L}_{\text{total}}(\theta, \psi) = -\mathcal{L}^{\text{CLIP}}(\theta) + c_v \, \mathbb{E}_t\big[(V_\psi(s_t) - \hat{R}_t)^2\big] - c_e \, \mathbb{E}_t\big[H(\pi_\theta(\cdot \mid s_t))\big], \tag{11}$$

where H is differential entropy for Gaussian policies. Minimizing $\mathcal{L}_{\text{total}}$ corresponds to ascending $\mathcal{L}^{\text{CLIP}}$ while fitting the critic and discouraging collapse of exploratory variance. Updates use mini-batches over several epochs on each rollout buffer; hyperparameters (learning rate, rollout length `n_steps`, batch size, γ, λ, ϵ, c_v, c_e) are set in YAML under `hexapod_training_new/configs/`.

3 Track I: Surrogate optimization and firmware translation (complete)

Status: done. This track implements Sec. 2.1: the parameterization of $\boldsymbol{\phi}$, sinusoidal `leg_commands()`, the eight-state surrogate ODE, the explicit cost (5), box-constrained L-BFGS-B with random restarts, and deterministic export of $\boldsymbol{\phi}^\star$ to firmware. It isolates gait timing and amplitude before closing any sensor loop on hardware.

Deliverables. `hexapod_no_imu_optimization.py` (L-BFGS-B, constraints, random restarts); `no_imu_optimization_summary.json`; `sync_gait_params_from_json.py`; `hexapod_openloop_sine.ino` with `optimized_gait_params.h`. Surrogate metrics and $\boldsymbol{\phi}^\star$ appear in Tables 6–7; the MuJoCo training model is illustrated in Fig. 4.

4 Track II: PPO with IMU-informed observations and preset tripod structure (near complete)

Status: near complete (simulation and training stack; hardware deployment partial). Here the policy is $\pi_\theta(a|s)$ with s containing **proprioception** (joint positions and velocities), **base kinematics** from the simulator, and—in IMU-focused configs—accelerometer and gyroscope readings (or their low-pass / complementary-filter equivalents) so that roll, pitch, and body rates enter the observation stack, matching the intended Walter firmware sensing. The **preset tripod structure** is enforced by one or more of: (i) a fixed gait clock or phase features in s; (ii) structured policies (e.g., sCPG or phase-parameterized heads) that bias action toward tripod alternation; or (iii) reward terms that penalize deviation from productive tripod coordination. PPO updates θ using Eqs. (8)–(10) on MuJoCo rollouts.

Relation to mathematics. The environment implements P and r; the agent maximizes $\eta(\theta)$ subject to PPO's trust-region-like clipping. IMU channels shrink the *partial observability* gap between simulation and the physical BMI270 stack; training runs such as `imu6_001` / `imu_bno10_001` in `hexapod_training_new/runs/` document this observation mode. Released evaluations (Table 8)

remain predominantly proprioceptive / path-following; extending hardware parity for Track II requires flashing a policy or residual layer while the Maestro executes tripod-compatible targets.

5 Track III: Preset tripod, IMU, and vision for distance-traveled PPO (planned)

Status: not done. The goal is to augment the state with **visual odometry** or pose estimates from an overhead or onboard camera so the agent (or a critic) can optimize **distance traveled** along a body or world axis with less simulator privilege. Let $p_t \in \mathbb{R}^2$ be horizontal position from integration of visual motion or fiducial tracking, and let $\Delta_t = \|p_t - p_0\|$ or forward displacement $x_t - x_0$. A natural reward increment is

$$r_t = \alpha\,(x_{t+1} - x_t) - \beta\,\|\omega_t\|^2 - \rho\,\mathrm{penalty}_{\mathrm{fall}} - \kappa\,\mathrm{TV}(a)_t, \tag{12}$$

with IMU rates ω_t penalizing harsh attitude changes, and optional terms for staying upright. The **observation** becomes $s_t = [s_t^{\mathrm{IMU/proprio}}, f_t]$ where f_t is a vector of visual features or estimated pose; PPO then applies the same surrogate (10) with expanded input dimension.

Open problems. Sim-to-real on vision (latency, calibration, lighting); defining r_t from *estimated* versus ground-truth displacement; and maintaining the preset tripod prior while learning residuals. This track is the logical successor once Track II policies are stable on Walter with IMU feedback.

Table 1: Summary of the three development tracks for *Walter*.

Track	Technical core	Status
I	Surrogate cost $\mathcal{J}(\boldsymbol{\phi})$ (5); L-BFGS-B; JSON→`.h`→ open-loop firmware	Complete
II	PPO $\mathcal{L}^{\mathrm{CLIP}}$ (10); GAE (8); IMU in s; tripod prior / structured policy	Near complete (simulation; hardware partial)
III	Same as II plus visual features or pose f_t; rewards from estimated distance / odometry	Not started

6 Technical Discussion

6.1 Biological and control context

The Introduction motivates *Walter* as a layered, bio-inspired architecture; this subsection records additional literature touchpoints used in detailed design. Decentralized feedback atop rhythmic pattern generators remains the standard modeling idiom for insect locomotion [4, 5]. Complementary or quaternion attitude filters address accelerometer–gyro fusion when magnetometers are untrust-

worthy [6, 8]; hexapod posture control on irregular terrain likewise foregrounds body orientation and angular velocity [7]. Disturbance-triggered gait transitions provide a classical benchmark for reflex and supervisory layers [9].

6.2 Layered architecture (specification)

The intended runtime stack, documented in `docs/ARCHITECTURE.md`, uses two timescales. A **fast tick** (∼10–20 ms) reads the IMU, fuses attitude, computes a stability score S, applies reflex adjustments, steps the gait interpolator, clamps commands, and streams twelve servo setpoints. A **slow tick** (each gait cycle) aggregates windowed RMS of S, runs adaptation, and saves the best parameter vector. The complementary blend

$$\alpha \in (0,1), \quad \theta_{\text{roll}} \leftarrow \alpha(\theta_{\text{roll}} + \omega_x \Delta t) + (1-\alpha)\,\theta_{\text{roll,accel}}, \tag{13}$$

(and analogously for pitch) is the nominal attitude update.

6.3 Open-loop sinusoidal gait (implemented)

For the no-IMU milestone, each leg follows resting abduction and flexion angles plus sinusoidal variation at common frequency f, with tripod group B offset by π from group A. This is the same law optimized in Sec. 2.1 and delivered in Track I (Sec. 3). Amplitudes follow default tables shared by Python and `optimized_gait_params.h`.

6.4 MuJoCo and reinforcement learning (simulation)

High-fidelity training uses Gymnasium + MuJoCo [1] with PPO [2] as implemented in Stable-Baselines3 [3]; the mathematical objective is Eq. (10) with advantages (8). Tracks II–III (Sec. 4–5) specialize the observation and reward; YAML configs set γ, λ, learning rate, network widths, PD gains, and terrain. A representative static render of the simulation model appears in Fig. 4 (Sec. 9.2).

6.5 Mechanical design: CAD and 3D-printed parts

Walter's chassis and legs are built from **parts designed in CAD** and produced by **additive manufacturing** (3D printing): a central body, six leg assemblies, and two-degree-of-freedom linkages per leg (abduction and flexion axes) that mate to the servo output splines. The design prioritizes a repeatable assembly for twelve actuated joints, clearance for wiring, and attachment of the electronics tray and battery. Figure 1 shows an overall render; Figure 2 shows the body, leg, and linkage components used in the assembly. Fasteners and servo hardware complete the mechanical

stack; material usage (filament mass) and printed-part cost are not included in the dollar totals in Tables 3–5.

6.6 Electronics, Hexapod V2 PCB, and safety

The bill of materials matches purchased components: Teensy-class MCU (SparkFun 16771), Pololu 1354 eighteen-channel driver, buck converter, BMI270 via Qwiic cabling, dual-cell LiPo and charger, terminals, and servos. Custom *Hexapod V2* PCBs (revision v5 dated 2026-04-03 on the latest order) integrate power and signal routing for Walter's stack; Figure 3 shows a board photo and schematic excerpt. `docs/FIRMWARE.md` records UART to the servo controller, channel ordering FL/FR/ML/MR/RL/RR with abduction then flexion per leg, and pulse mapping $u_k = \mathrm{clamp}(c_k + d_k q_k)$ (typical bounds 600–2400 μs). Open-loop firmware applies degree-to-command gains and clamps before streaming compact-protocol targets; these bench gains are not outputs of the reduced-order optimizer.

7 Requirements and Constraints

7.1 Source of requirements

Engineering requirements are traced to phase exit criteria in `docs/PHASES.md` (Phases 1–4: IMU telemetry, reflexes, cautious mode and recovery, online adaptation), supplemented by safety and interface constraints in `docs/FIRMWARE.md` and the open-loop scope in `firmware/hexapod_openloop_sine/README.m` `PHASES.md` serves as the team's consolidated requirements-and-constraints specification for firmware behavior.

7.2 Summary: met vs. not met

Met (repository evidence): discrete tripod baseline (`hexapod_walking_v2`); open-loop optimizer and summary JSON/CSV; firmware `hexapod_openloop_sine` with JSON-to-header script; calibration helpers; design, architecture, and firmware documentation; CAD/3D-printed mechanical design for *Walter* documented in this report (Figs. 1–2); MuJoCo/PPO training and evaluation artifacts under `hexapod_training_new/` and `e90_repo/`.

Partial / not in version control: Phase 1 IMU telemetry sketch path may be absent as `hexapod_imu_test/`; Phases 2–4 adaptive firmware (`hexapod_adaptive/`) not present; full on-device learning as described in `docs/on_device_learning/` remains planning relative to the checked-in firmware tree.

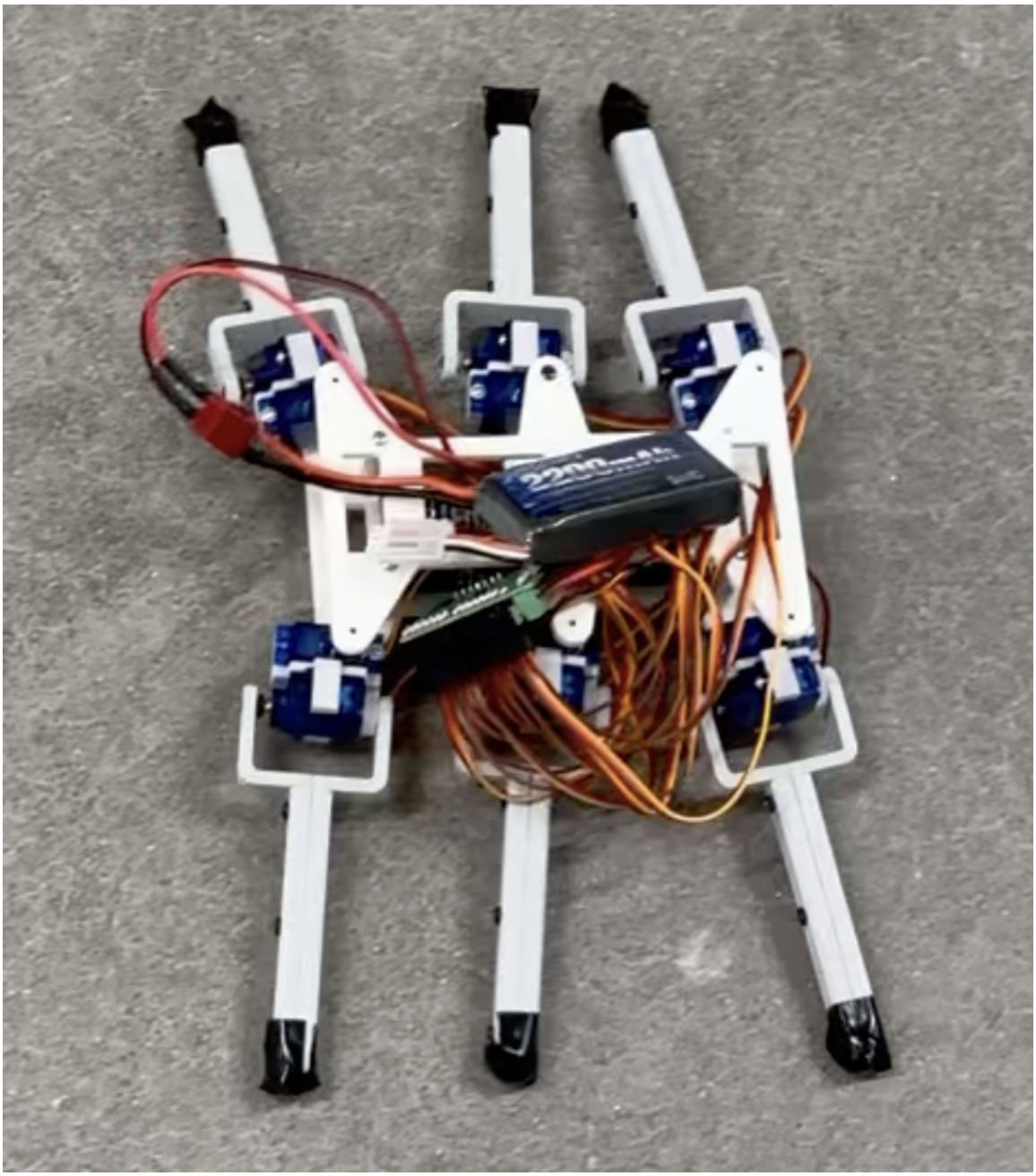
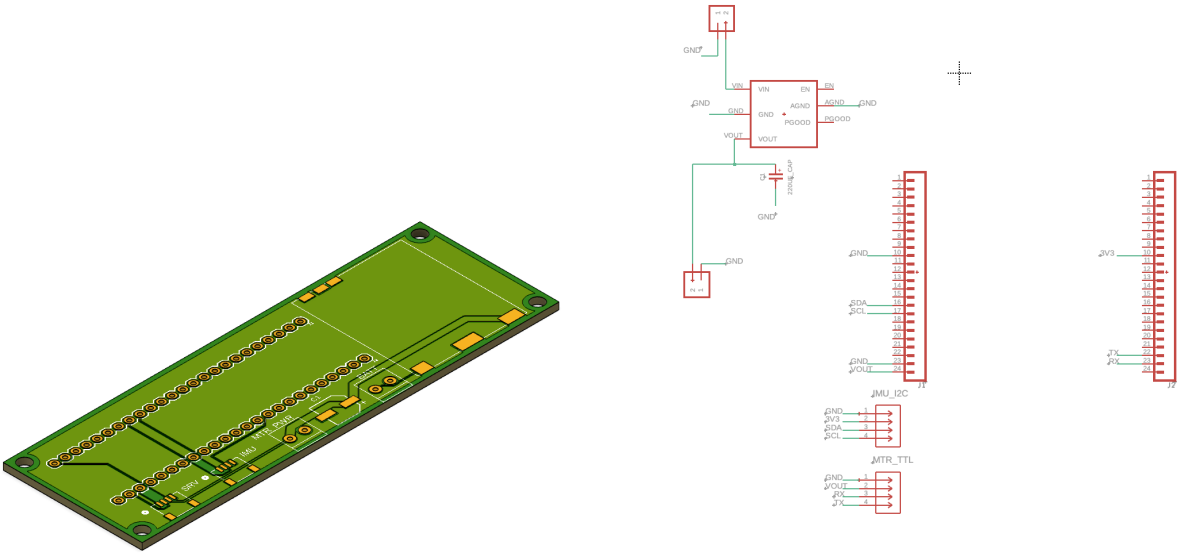


Figure 1: Walter: full-assembly render of the twelve-DOF hexapod (CAD; Walking Alternating Tripod for Evolutionary Research).

(a) Body (b) Leg assemblies (c) Linkage

Figure 2: CAD-designed 3D-printed subassemblies for Walter: chassis, legs, and per-leg linkage.



(a) Assembled PCB (Hexapod V2) (b) Schematic (excerpt)

Figure 3: Custom electronics for Walter: Hexapod V2 board and schematic.

Table 2: Requirement traceability. Evidence points to repository artifacts.

ID	Requirement	Status	Evidence / notes
R1	Phase 1: IMU read, complementary filter, USB telemetry ∼50 Hz; gait unchanged	Not met (in repo)	`PHASES.md`; folder may be missing
R2	Phase 2: postural reflexes, clamps, serial tuning, EEPROM	Not met (in repo)	`ARCHITECTURE.md` pseudocode
R3	Phase 3: stability score, cautious mode, push recovery, fall freeze	Not met (in repo)	Documented only
R4	Phase 4: five-parameter adaptation, cost J, SPSA/bandit, EEPROM	Not met (in repo)	Documented only
R5	Hardware: MCU + eighteen-channel driver + servos + documented map + PCBs + Walter mechanical CAD/3D parts	Met	BOM, Figs. 1–3, `FIRMWARE.md`
R6	Safety: pulse clamps; parameter slew (adaptive layer)	Partial	Clamps in v2/openloop; slew in spec
R7	Open-loop: optimizer/firmware parity; JSON export	Met	Optimizer, `sync_gait_params_from_json.py`
R8	Simulation: MuJoCo env, PPO training, documented evals	Met	`hexapod_training_new/`, `e90_repo/released_checkpoints/`

7.3 Project cost

Tables 3–4 summarize **purchased** electronics (three identical platforms) and **fabrication** costs for three PCB/stencil orders. Table 5 gives the grand total. Servo counts in Table 3 are per platform (eight units listed per BOM line); completing twelve DOF per robot may require additional servos beyond those lines if not already stocked.

8 Methods

8.1 Mechanical assembly (*Walter*)

Printed body and leg components are fastened per the CAD design; each leg receives two servos driving abduction and flexion through the linkage. The custom PCB mounts to the chassis and connects to the Pololu driver and Teensy; power and ground respect the buck converter and screw terminals from the BOM. Bench alignment uses `set_zero` and side-only test sketches before closed-loop or open-loop gait tests.

Table 3: Electronics bill of materials, **one platform** (USD). The team purchased three identical sets.

Part	Description	Vendor	Qty	Unit	Line
SparkFun 16771	Microcontroller	DigiKey	1	37.20	37.20
Pololu 1354	18-channel servo driver	DigiKey	1	46.95	46.95
DFRobot DFR1202	Step-down buck converter	DFRobot	1	12.90	12.90
SparkFun BMI270	6-DoF IMU	DigiKey	1	18.50	18.50
SparkFun 11855	2×7.4 V LiPo + charger	Amazon	1	27.99	27.99
SparkFun 14417	Female Qwiic connector	DigiKey	2	0.60	1.20
SparkFun 17259	100 mm Qwiic cable	DigiKey	1	1.95	1.95
SparkFun 17261	Qwiic cable to female header	DigiKey	1	1.95	1.95
Amphenol YO0201500000G	1×2 screw-down terminal	DigiKey	2	1.28	2.56
DFRobot SER0039	Servo motor	DigiKey	8	5.90	47.20
Subtotal	**one platform**				**198.40**
Electronics total	**three platforms (×3)**				**595.20**

Table 4: PCB and stencil orders, *Hexapod V2* (USD order totals including merchandise, shipping, duties/taxes, and sales tax where charged by the vendor).

Order ID	Description / notes	Status	Total
Y4-11911448A	PCB prototype (5 pcs) + SMT stencil; v5_2026-04-03	Awaiting payment	74.24
W2026030804205410	PCB + stencil shipment	Shipped	68.26
W2026021709554129	PCB + stencil (earlier order)	Shipped	54.97
PCB subtotal			**197.47**

Table 5: Grand total project cost (USD) for reported purchases and fabrication.

Category	Amount
Electronics (three platforms)	595.20
PCB fabrication and stencils (three orders)	197.47
Grand total	**792.67**

8.2 Baseline discrete gait

`firmware/hexapod_walking_v2/hexapod_walking_v2.ino` implements a deterministic tripod state machine with smoothed keyframes (`moveSmoothTo`).

8.3 Reduced-order simulation optimization

`hexapod_brain_roadmap copy/ode_optimization_no_imu/hexapod_no_imu_optimization.py` integrates the surrogate model with `scipy.integrate.solve_ivp`, evaluates the objective on a steady-state window, and calls `scipy.optimize.minimize` (L-BFGS-B). Outputs include `no_imu_optimization_s` and optional CSV/plots.

8.4 Parameter sync and open-loop firmware

From `firmware/hexapod_openloop_sine/`, run `python3 sync_gait_params_from_json.py` to regenerate `optimized_gait_params.h`. The sketch evaluates the sinusoidal law at `INTERP_DELAY_MS` (default 10 ms), applies startup frequency ramping, maps degrees to integer commands, clamps, and sends Pololu compact-protocol targets on `Serial1`.

8.5 Bench testing and calibration

`firmware/set_zero/` and side-only test sketches support calibration. The open-loop README documents sign and scale tuning when feet drag or the robot shuffles backward.

8.6 MuJoCo PPO training and evaluation

Training uses `hexapod_training_new/train.py` with YAML configs under `configs/`; runs write `training_summary.yaml`, checkpoints, and logs under `runs/<run_name>/`. Evaluation uses `evaluate.py` with the same config as training; the `e90_repo/released_checkpoints/` tree archives representative `.zip` weights and `evaluations/<name>/REPORT.md` summaries for slides and reporting.

9 Results

9.1 Reduced-order optimization (surrogate model)

Table 6 summarizes `hexapod_brain_roadmap copy/ode_optimization_no_imu/no_imu_optimization_summary` These metrics characterize the *surrogate* model; hardware performance requires bench validation of gains, centers, and directions.

Table 6: Optimizer-reported metrics (committed summary JSON).

Quantity	Value
Gait frequency f	1.80 Hz
Mean forward speed (steady window)	0.234 m/s
Total displacement (rollout)	1.86 m
Roll RMS	0.440 rad
Pitch RMS	~ 0 (near zero in file)
Vertical position RMS	0.086 m
Peak servo usage ratio (proxy)	0.152
Contact balance std. dev.	0.388
Dominant oscillation frequency (estimated)	3.54 Hz
Objective cost $\mathcal{J}(\boldsymbol{\phi}^\star)$	-228.38

Table 7 lists the corresponding optimized parameter vector $\boldsymbol{\phi}^\star$ (rest angles, phase shifts, frequency, and swing lift) exported to `optimized_gait_params.h`. Flexion phase shifts sit at the lower bound ($-90°$), trading stride phasing for the surrogate's forward-speed and regularization terms; hardware tuning may move off that bound.

Table 7: Optimized decision variables $\boldsymbol{\phi}^\star$ from L-BFGS-B (same JSON as Table 6). Angles in degrees; leg order FL, FR, ML, MR, RL, RR.

Quantity	Value
Gait frequency f	1.800 Hz
Abduction rest $\boldsymbol{\theta}_{\mathrm{abd,rest}}$	(10, 10, 0, 0, -10, -10)
Flexion rest $\boldsymbol{\theta}_{\mathrm{flx,rest}}$	(32.00, 32.00, 34.00, 34.00, 32.00, 32.00)
Abduction phase $\boldsymbol{\delta}_{\mathrm{abd}}$	$\approx \mathbf{0}$ (magnitude $< 10^{-6}$ deg)
Flexion phase $\boldsymbol{\delta}_{\mathrm{flx}}$	(-90, -90, -90, -90, -90, -90)
Swing lift amplitude	12.0 deg (`lift_swing_deg_peak`)

9.2 MuJoCo simulation (high-fidelity model)

Track II–III training uses Gymnasium with the MuJoCo asset `hexapod_training_new/assets/hexapod.xml`: checkerboard ground, twelve torque-controlled joints, and RK4 integration at 0.002 s (see configs for frame skipping). Figure 4 is an **offscreen MuJoCo render** (software `Renderer`, no interactive viewer): the model was advanced under gravity with zero actuator commands until the body settled on the plane, then a fixed camera was used. The view documents the mesh, materials, and lighting used for PPO rollouts; it is not a policy-controlled frame. Interactive inspection is available via `scripts/live_mujoco_viewer.py` with a trained checkpoint.

Figure 4: MuJoCo screenshot of the Walter hexapod model used for PPO training and evaluation (post-settle, zero control; offscreen render).

9.3 MuJoCo PPO evaluations (released checkpoints)

Table 8 quotes short evaluation protocols documented in `e90_repo/released_checkpoints/evaluations/`. Rewards are **not** comparable across configs (different shaping and scaling); forward displacement and path length provide a common physical readout within each experiment family.

Table 8: Representative PPO evaluation means (300 env steps per episode unless noted; see each REPORT.md).

Checkpoint / run	Task / policy	Fwd. $+x$	2D path len.	Note
`mlp_clean_run005_best`	Circle path; MLP 256×256 PPO; `mlp_run_005` (301k steps)	0.18 m	0.27 m	Mean reward ≈ 680
`terrain_low_shake_run001_best`	Flat / rough / steps; MLP PPO	0.33 m mean over 3 ep.	—	Flat/rough ≈0.47 m fwd.; steps harde

The `pd_velocity_window_run018_best` checkpoint documents an sCPG + PD + velocity-window experiment; its released short eval shows small net displacement—useful as an archival structured-policy datapoint rather than a primary locomotion benchmark (see that folder's `REPORT.md`).

9.4 Requirements and cost (results view)

Open-loop, hardware documentation, PCB procurement, and simulation requirements (R5–R8) are supported by artifacts. Phase 1–4 adaptive firmware requirements (R1–R4) are **not** satisfied in the repository snapshot. Purchased engineering spend for this accounting is **US$792.67** (Tables 3–5).

10 Discussion

Context. Track I gives an interpretable, verifiable optimization-to-code pipeline. Track II applies the PPO mathematics of Sec. 2 to rich MuJoCo dynamics with IMU-capable observation stacks and tripod-structured policies; Track III would add visual state for distance-centric rewards but is not yet built. We separate **simulation** results from **hardware** validation: closing the loop on II–III requires IMU integration, possible camera/odometry, power-under-load tests, and honest reporting against `PHASES.md` exit criteria.

What went well. A named platform (*Walter*) with traceable CAD and 3D-printed parts; shared command laws between Python and C++; JSON-to-header automation; a stable servo-driver and custom PCB stack; a growing library of RL configs, runs, and released evaluations.

Limitations. Model mismatch and servo saturation can invalidate predicted speeds; manual degree-to-pulse scaling; open-loop policies do not reject disturbances. RL rewards are task-specific—cross-comparing raw reward across configs is misleading.

Lessons for future E90 students. Freeze a logging format and baseline gait before adding sensors; budget time for bench tuning after simulation; document PCB spins and BOM multiples early; keep a written trace from optimizer or checkpoint to deployed constants.

11 Conclusion

We presented *Walter* with CAD/PCB/cost documentation; a mathematical treatment of surrogate optimization (L-BFGS-B), MDPs, GAE, and PPO; and **three tracks**: (1) optimization-to-code (*complete*); (2) PPO with IMU and preset tripod structure (*near complete* in sim); (3) IMU + vision + distance PPO (*planned*). Materials are at https://github.com/HowardWHSrun/WALTER. Major takeaways: interpretable optimization complements learned policies; IMU and eventually vision shrink the sim-to-real observation gap; requirement tracing must distinguish which track is deployed on hardware.

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A Public repository

The public project URL is https://github.com/HowardWHSrun/WALTER. Appendix paths below refer to the local Engineering 90 workspace layout at the time of writing; the GitHub repository is intended to subsume or mirror these components for reproducibility.

B Repository map (selected local paths)

- **Adaptive firmware and docs:** `RL Temp/adaptive_ondevice_hexapod/README.md`, `docs/DESIGN.md`, `docs/ARCHITECTURE.md`, `docs/PHASES.md`, `docs/FIRMWARE.md`
- **Firmware sketches:** `firmware/hexapod_walking_v2/`, `firmware/hexapod_openloop_sine/`, `firmware/set_zero/`, side-only tests
- **Reduced-order optimization:** `hexapod_brain_roadmap copy/ode_optimization_no_imu/`
- **MuJoCo PPO training:** `RL Temp/hexapod_training_new/` (`train.py`, `configs/`, `runs/`)
- **Released checkpoints and evals:** `RL Temp/e90_repo/released_checkpoints/`
- **Presentation notes:** `Presentation/E90_MidSemester_SpeakerScript.tex` (and PDF)
- **This folder (figures):** `walter_assembly.png`, `Body.png`, `Legs.png`, `Linkage.png`, `pcb_board.png`, `pcb_schematic.png`, `mujoco_hexapod_sim.png` (plus originals with longer filenames)